

$$1. - V(S_0) = 100 \quad V(S_1) = 20$$

$$P^{aG} = \begin{bmatrix} 0.3 & 0.7 \\ 0.1 & 0.9 \end{bmatrix} \quad P^{aB} = \begin{bmatrix} 0.9 & 0.1 \\ 0.7 & 0.3 \end{bmatrix}$$

$$c(\lambda, a_G) = 10 \quad \gamma = 0.9$$

$$c(S_0, a_B) = 50$$

$$c(S_1, a_B) = 0$$

$$Q(\lambda, a) = C(\lambda, a) + \gamma \sum_n P(\lambda' | \lambda, a) \cdot V(\lambda')$$

$$Q(S_1, a^G) = 10 + 0.9(0.1 \times V(S_0) + 0.9 \times V(S_1)) = 35.2$$

$$\downarrow \quad \downarrow$$

$$P(S_0 | S_1, a^G) \quad P(S_1 | S_1, a^G)$$

$$Q(S_1, a^B) = 0 + 0.9(0.7 \times 100 + 0.3 \times 20) = 68.4$$

$$\text{Optimal action} \rightarrow \min(Q(S_1, a^G), Q(S_1, a^B)) = 35.2 \rightarrow \boxed{a^G}$$

$$2. - V^\pi = C^\pi + \gamma P^\pi V^\pi \Leftrightarrow (I - \gamma P^\pi) V^\pi = C^\pi \quad (\pi \text{ always chooses } a^G)$$

$$C^\pi = \begin{bmatrix} 10 \\ 10 \end{bmatrix} \quad (c(S_0, a^G) = c(S_1, a^G) = 10) \quad P^\pi = P^{a^G}$$

$$V(S_0) = 10 + 0.9 \cdot [0.3 \cdot V(S_0) + 0.7 \cdot V(S_1)] \Leftrightarrow V(S_0) = 0.73 V(S_0) - 0.63 V(S_1) = 10$$

$$V(S_1) = 10 + 0.9 \cdot [0.1 \cdot V(S_0) + 0.9 \cdot V(S_1)] \Leftrightarrow -0.09 V(S_0) + 0.19 V(S_1) = 10$$

~~$$0.73 V(S_0) - 0.63 V(S_1) = 10$$~~

$$-0.09 V(S_0) + 0.19 V(S_1) = 10 \Leftrightarrow V(S_1) = \frac{10 + 0.09 V(S_0)}{0.19}$$

$$0.73 V(S_0) - 0.63 V(S_1) = 10 \Leftrightarrow 0.73 V(S_0) - 0.63 \left(\frac{10 + 0.09 V(S_0)}{0.19} \right) = 10$$

$$(\text{multiply by } 0.19) \Leftrightarrow 0.1387 V(S_0) - 0.63(10 + 0.09 V(S_0)) = 1.9$$

$$\Leftrightarrow 0.1387 V(S_0) - 6.3 - 0.0567 V(S_0) = 1.9$$

$$\Leftrightarrow 0.082 V(S_0) = 8.2 \Leftrightarrow \boxed{V(S_0) = 100}$$

$$V(S_1) = \frac{10 + 9}{0.19} = 100$$

$$\boxed{V(S_0) = V(S_1) = 100}$$

③

3.1)

$$V_{\text{new}}(s) = V_{\text{old}}(s) + C$$

$$\begin{aligned} TV_{\text{new}}(s) &= \min_a \left[C(s, a) + \gamma \sum_{s'} P(s' | s, a) V_{\text{new}}(s') \right] \\ &= \min_a \left[C(s, a) + \gamma \sum_{s'} P(s' | s, a) (V_{\text{old}}(s') + C) \right] \\ &= \min_a \left[C(s, a) + \gamma \sum_{s'} P(s' | s, a) V_{\text{old}}(s') + \gamma \left(\sum_{s'} P(s' | s, a) \right) C \right] \\ &\quad \underbrace{\qquad\qquad\qquad}_{=1, \text{ since the sum of all probabilities is 1}} \\ &\stackrel{\substack{\text{does not} \\ \text{depend on } a}}{\gamma C} = \min_a \left[C(s, a) + \gamma \sum_{s'} P(s' | s, a) V_{\text{old}}(s') \right] + \gamma C \end{aligned}$$

$$TV_{\text{new}}(s) = TV_{\text{old}}(s) + \gamma C \quad //$$

3.2.

Take any two value functions V_1 and V_2 . Define $C = \|V_1 - V_2\|_{\infty}$, meaning $|V_1(s) - V_2(s)| \leq C$ for all s , which means:

$$V_1 - C \leq V_2 \leq V_1 + C$$

By the monotonicity of T and 3.1, applying T to both sides gives:

$$\begin{aligned} TV_1 - \gamma C &\leq TV_2 \leq TV_1 + \gamma C \Rightarrow \\ \Rightarrow \|TV_1 - TV_2\|_{\infty} &\leq \gamma C = \gamma \|V_1 - V_2\|_{\infty} \end{aligned}$$

Since $\gamma = 0.9 < 1$, applying T shrinks the distance between any two value functions by a factor of γ . By the Banach Fixed Point Theorem, any contraction mapping on a complete metric space has exactly one fixed point, and repeated application converges to it.

This means no matter what V_0 you start with, the sequence $V, TV, T^2V, \dots$ converges to the same unique V^* , the optimal value function.